



Real-time Deployment

LECTURE

Databricks Model Serving

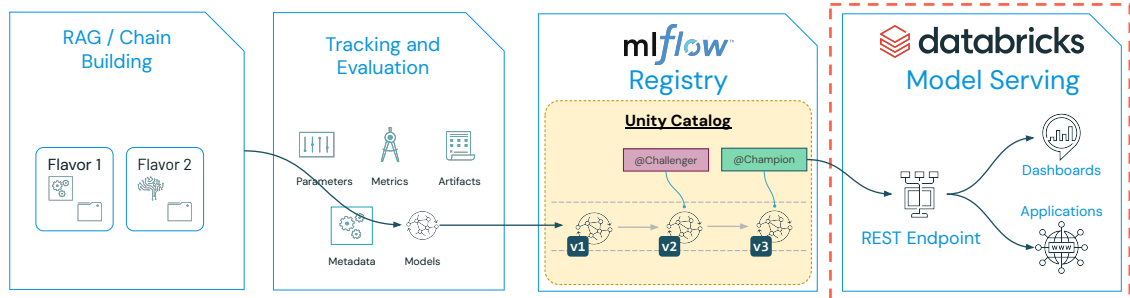


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In this lecture Databricks Model Serving, we explore the MLflow development lifecycle and the features of Databricks Model Serving.

MLflow and Development Lifecycle

MLflow integration to seamlessly deploy models



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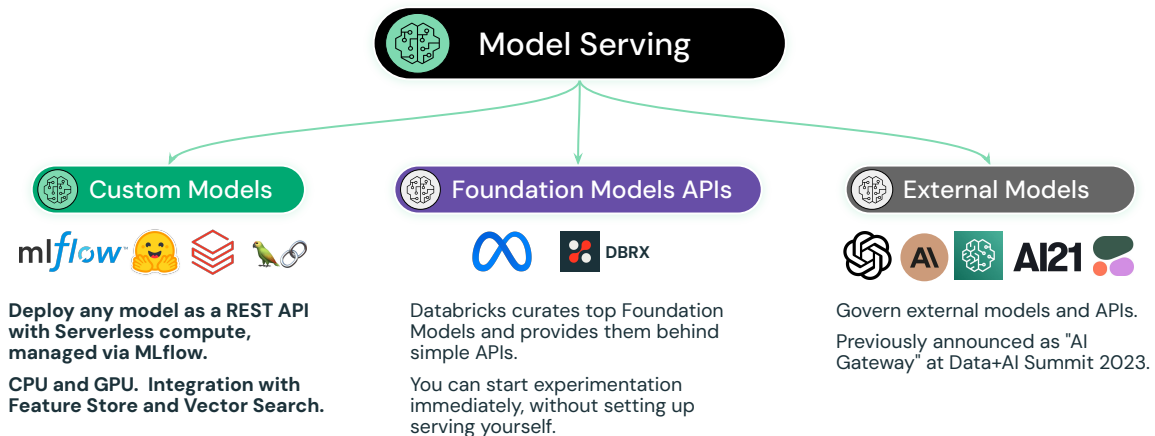
What does this integration with MLflow look like?

Here's a high level picture of a typical lifecycle of an ML model.

In a typical ML model lifecycle, Model Serving is tightly integrated with the Model Registry. The Model Registry allows efficient management and staging of models, while Model Serving facilitates deployment of any version of the model through REST endpoints, providing flexibility and ease of use for real-time applications.

Databricks Model Serving

Unified UI, API & SDK for managing all types of AI Models



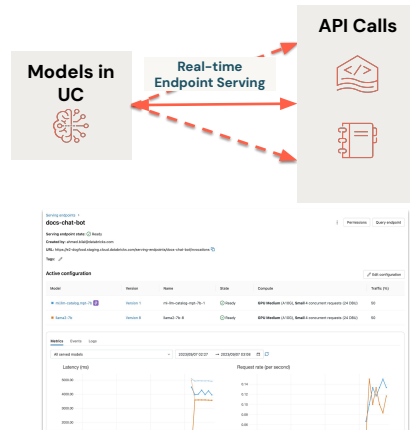
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Given that there are so many options available—whether you want to build your own models, use pre-existing open source foundation models, or even tap into proprietary models—all of these can be accessed on the Databricks platform through a unified API or SDK. This setup makes it really simple to switch between models behind the scenes without needing to rewrite either the backend or even the frontend of your system. Essentially, the way you interact with the system remains the same. On top of that, this unified API provides a lot of built-in functionality for monitoring your system right out of the box.

Databricks Model Serving

Deploy registered models behind real-time, production-grade APIs

- Production-ready, serverless solution to simplify real-time ML model deployment.
- Deploy models as an API to integrate model predictions with applications or websites.
- Built-in payload logging and infrastructure/system observability.



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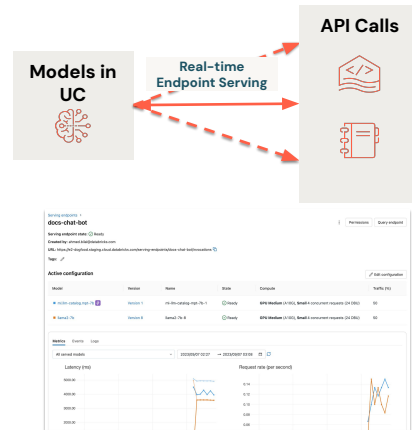
We offer built-in payload logging, so every time a request comes into the endpoint, it's automatically recorded in a Delta table for ad hoc analysis. In addition to that, infrastructure metrics are also available and can be accessed through the API.

Databricks Model Serving

Deploy registered models behind real-time, production-grade APIs

Benefits:

- Reduces operational costs
- Streamlines the ML lifecycle
- Enables Data Science teams to focus on the core task of integrating production-grade real-time ML into their solutions
- Autoscaling and serverless compute
- Works with custom, foundational, and external models
- Built-in A/B testing



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To summarize the value of Databricks Model Serving, we look at the core benefits it provides when deploying production-grade, real-time APIs.

First, it significantly reduces operational costs and streamlines the ML lifecycle. By handling the complex infrastructure and scaling automatically, it enables Data Science teams to concentrate on their core task: integrating production-grade real-time ML into their solutions.

The platform offers autoscaling and serverless compute, which helps manage the expense and performance challenges we discussed earlier.

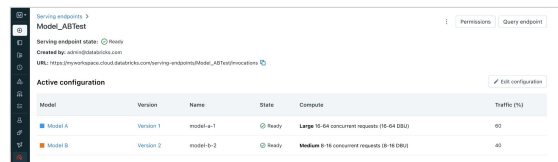
Importantly, it provides flexibility by working seamlessly with custom models, Databricks foundational models, and external models.

Finally, it includes built-in A/B testing, which is critical for continuously improving and validating your generative AI systems.

Databricks Model Serving

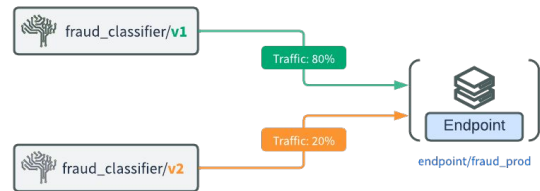
Online evaluation

- Supports online evaluation strategies such as A/B testing or canary deployments through the ability to serve multiple models to a serving endpoint



The screenshot shows the 'Active configuration' for a serving endpoint. It includes a table with columns: Model, Version, Name, State, Compute, and Traffic (%).

Model	Version	Name	State	Compute	Traffic (%)
Model A	Version 1	model-a-1	Ready	Large 10-64 concurrent requests (10-64 DBU)	80
Model B	Version 2	model-b-2	Ready	Medium 8-16 concurrent requests (8-16 DBU)	20



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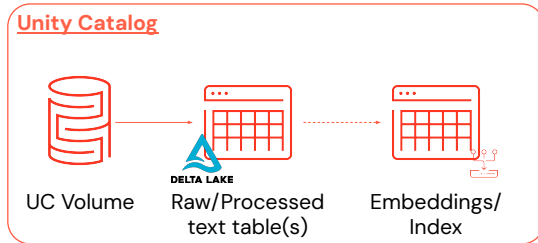
The built-in A/B testing capability mentioned on the previous slide is a key feature for online evaluation.

Databricks Model Serving supports sophisticated online evaluation strategies. This includes techniques like A/B testing or canary deployments.

The way this is achieved is through the platform's ability to serve multiple models to a single serving endpoint. For instance, you could route 80% of live traffic to the currently established 'Champion' Model Version 1, and the remaining 20% to a new 'Challenger' Model Version 2 to test its performance in a real-time environment before a full rollout. This ensures that any model promotion is backed by real production metrics.

Databricks Model Serving

Inference tables

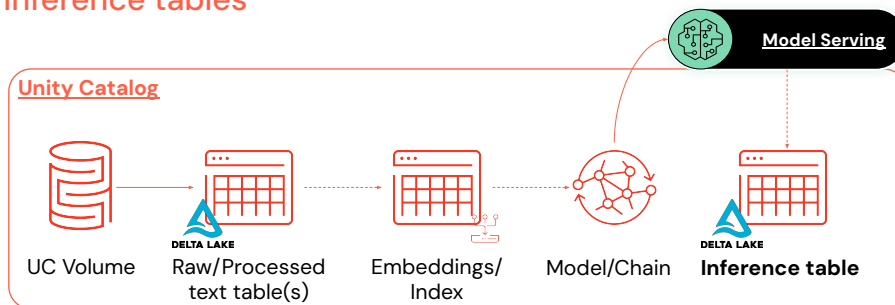


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To give you an idea of how monitoring works for a deployed generative AI system within Databricks' serving infrastructure, there's a key feature called the inference table from model serving. This feature allows for effective online monitoring of the system in real time.

Databricks Model Serving

Inference tables



Inference tables for monitoring and debugging models:

Each [request-response] is appended to a delta table in Unity Catalog

- Use as retraining/fine-tuning dataset for next iteration of your model(s)
- Perform diagnostics and debugging of suspicious inferences
- Create a dataset of mislabels data to be re-label



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As the name suggests, the inference table is simply a Databricks table created within a chosen catalog and schema, designed to log every request, response, and extra metadata related to a model deployed on a serving endpoint. To break it down: you have a model in your catalog, deploy it using the model serving infrastructure, and once payload logging is enabled, any incoming request to that endpoint is automatically captured in a Delta table called the inference table.

This not only lets you use real-world data to retrain or fine-tune your models in the future but also helps with online diagnosis, debugging, and system monitoring—measuring things like model drift or spikes in toxicity or perplexity. Moreover, it allows you to create sets of mislabeled or relabel-needed data. For example, if a model's response isn't accurate, those instances can be flagged in another table and later used to retrain the model, ensuring it better handles edge cases and remains well-aligned with intended outcomes.



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